

Program 21: Swap Case in DataFrame

AIM

To write a Pandas program to swap the cases of a specified character column in a given DataFrame.

CODE

```
import pandas as pd

# Sample DataFrame
df = pd.DataFrame({
    'Company': ['google', 'microsoft', 'APPLE', 'amazon'],
    'Employees': [1000, 2000, 3000, 4000]
})

print("Original DataFrame:")
print(df)

# Swapping cases of 'Company' column
df['Company'] = df['Company'].str.swapcase()

print("\nDataFrame after swapping cases:")
print(df)
```

OUTPUT

```
Original DataFrame:
   Company  Employees
0   google      1000
1 microsoft      2000
2   APPLE      3000
3  amazon      4000

DataFrame after swapping cases:
   Company  Employees
0  GOOGLE      1000
1 MICROSOFT      2000
2   apple      3000
3  AMAZON      4000
```

Result: The Pandas program successfully swapped the text cases of the 'Company' column.

Program 22: Draw a Labeled Line Chart

AIM

To write a Python program to draw a line with a suitable label on the x-axis, y-axis, and a title.

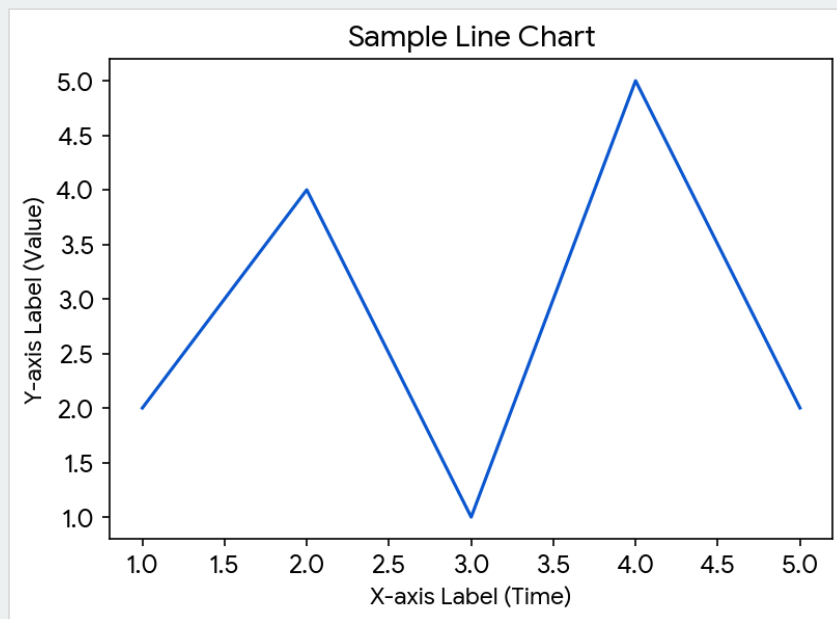
CODE

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [2, 4, 1, 5, 2]

plt.plot(x, y)
plt.xlabel('X-axis Label (Time)')
plt.ylabel('Y-axis Label (Value)')
plt.title('Sample Line Chart')
plt.show()
```

OUTPUT



Result: The Python program successfully plotted the line chart with the specified labels and title.